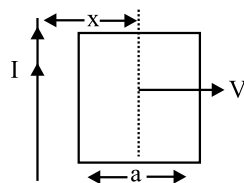


CHAPTER 6

Electromagnetic Induction

Magnetic Flux, Faraday's Law of E.M.F.

- The net magnetic flux through any closed surface is: (2023)
a. Negative b. Zero c. Positive d. Infinity
- An emf is generated by a generator having 100 turn coil, of loop area 1 m^2 . The coil rotates at a speed of one revolution per second and placed in a uniform magnet of 0.05 T perpendicular to the axis of rotation of the coil. The maximum value of emf is: (2023-Manipur)
a. 3.14 V b. 31.4 V c. 62.8 V d. 6.28 V
- The magnetic flux linked to a circular coil of radius R is:
 $\phi = 2t^3 + 4t^2 + 2t + 5 \text{ Wb}$
The magnitude of induced emf in the coil at $t = 5 \text{ s}$ is: (2022 Re)
a. 192 V b. 108 V c. 197 V d. 150 V
- A square loop of side 1 m and resistance 1Ω is placed in a magnetic field of 0.5 T . If the plane of loop of **perpendicular** to the direction of a magnetic field, the magnetic flux through the loop is: (2022)
a. zero weber b. 2 weber
c. 0.5 weber d. 1 weber
- The magnetic flux linked with a coil (in Wb) is given by the equation
 $\phi = 5t^2 + 3t + 16$
The magnitude of induced emf in the coil at the **fourth second** will be (2020-Covid)
a. 43 V b. 108 V c. 10 V d. 33 V
- A 800 turn coil of effective area 0.05 m^2 is kept **perpendicular** to a magnetic field of $5 \times 10^{-5} \text{ T}$. When the plane of the coil is rotated by 90° around any of its coplanar axis in 0.1 s , the emf induced in the coil will be: (2019)
a. 2 V
b. 0.2 V
c. $2 \times 10^{-3} \text{ V}$
d. 0.02 V
- A conducting square frame of side a and a long straight wire carrying current I are located in the same plane as shown in the figure. The frame moves to the right with a constant velocity V . The emf induced in the frame will be proportional to: (2015)



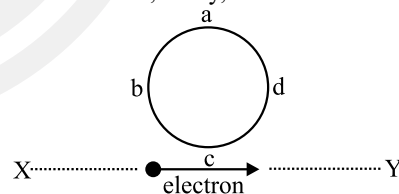
- $\frac{1}{(2x-a)^2}$
- $\frac{1}{(2x-a)(2x+a)}$
- $\frac{1}{(2x+2a)(2x+a)}$
- $\frac{1}{x^2}$

Lenz's Law

- (2024)

In the above diagram, a strong bar magnet is moving towards solenoid-2 from solenoid-1. The direction of induced current in solenoid-1 and that in solenoid-2, respectively, are through the directions:

- AB and CD
 - BA and DC
 - AB and DC
 - BA and CD
- An electron moves on a straight line path XY as shown. The $abcd$ is a coil adjacent to the path of electron. What will be the direction of current, if any, induced in the coil? (2015 Pre)



- No current induced
 - abcd
 - adcd
 - The current will reverse its direction as the electron goes past the coil
- A wire loop is rotated in a magnetic field. The frequency of change of direction of the induced e.m.f. is: (2013)
a. Six times per revolution b. Once per revolution
c. Twice per revolution d. Four times per revolution

Motional E.M.F.

- A big circular coil of 100 turns and average radius 10 m is rotating about its horizontal diameter at 2 rad s^{-1} . If the vertical component of earth's magnetic field at that place is $2 \times 10^{-5} \text{ T}$ and electrical **resistance** of the coil is 12.56Ω , then the maximum induced current in the coil will be: (2022)
a. 2 A b. 0.25 A c. 1.5 A d. 1 A

12. A wheel with 20 metallic spokes each 1 m long is rotated with a speed of 120 rpm in a plane perpendicular to a magnetic field of 0.4 G. The induced emf between the axle and rim of the wheel will be. (1 G = 10^{-4} T) (2020-Covid)

a. 2.51×10^{-5} V b. 4.0×10^{-5} V
c. 2.51 V d. 2.51×10^{-4} V

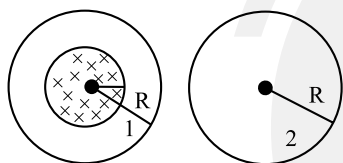
13. A metallic rod of mass per unit length 0.5 kg m^{-1} is lying horizontally on a smooth inclined plane which makes an angle of 30° with the horizontal. The rod is not allowed to slide down by flowing a current through it when a magnetic field of induction 0.25 T is acting on it in the vertical direction. The current flowing in the rod to keep it stationary is

(2018)

a. 14.76 A b. 5.98 A c. 7.14 A d. 11.32 A

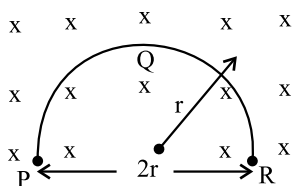
14. A uniform magnetic field is restricted within a region of radius r . The magnetic field changes with time at a rate $\frac{d\vec{B}}{dt}$. Loop 1 of radius $R > r$ encloses the region r and loop

2 of radius R is outside the region of magnetic field as shown in the figure below. Then the e.m.f generated is: (2016 - II)



- a. $-\frac{d\vec{B}}{dt}\pi R^2$ in loop 1 and zero in loop 2
b. $-\frac{d\vec{B}}{dt}\pi r^2$ in loop 1 and zero in loop 2
c. Zero in loop 1 and zero in loop 2
d. $-\frac{d\vec{B}}{dt}\pi r^2$ in loop 1 and $-\frac{d\vec{B}}{dt}\pi R^2$ in loop 2

15. A thin semicircular conducting ring (PQR) of radius r is falling with its plane vertical in a horizontal magnetic field B , as shown in figure. The potential difference developed across the ring when its speed is v , is: (2014)



- a. Zero
b. $Bv\pi r^2/2$ and P is at higher potential
c. $\pi r Bv$ and R is at higher potential
d. $2r Bv$ and R is at higher potential

Eddy Currents

16. In which of the following devices, the eddy current effect is **not** used? (2019)

a. Induction furnace b. Magnetic braking in train
c. Electromagnet d. Electric heater

Self Induction

17. Let us consider two solenoids A and B , made from same magnetic material of relative permeability μ_r and equal area of cross-section. Length of A is twice that of B and the number of turns per unit length in A is half that of B . The ratio of self inductances of the two solenoids, $L_A : L_B$ is (2024 Re)

a. 1 : 2 b. 2 : 1 c. 8 : 1 d. 1 : 8

18. A long solenoid has 1000 turns. When a current of 4 A flows through it, the magnetic flux linked with each turn of the solenoid is 4×10^{-3} Wb. The self inductance of the solenoid is: (2016 - I)

a. 4 H b. 3 H c. 2 H d. 1 H

Energy Stored or Work Done in Inductor

19. The magnetic potential energy stored in a certain inductor is 25 mJ, when the current in the inductor is 60 mA. This inductor is of inductance: (2018)

a. 1.389 H b. 138.88 H c. 0.138 H d. 13.89 H

Mutual Induction

20. Two conducting circular loops of radii R_1 and R_2 are placed in the same plane with their centres coinciding. If $R_1 > R_2$, the mutual inductance M between them will be **directly proportional** to: (2021)

a. $\frac{R_2}{R_1}$ b. $\frac{R_1^2}{R_2}$ c. $\frac{R_2^2}{R_1}$ d. $\frac{R_1}{R_2}$

21. A long solenoid of diameter 0.1 m has 2×10^4 turns per metre. At the centre of solenoid, a coil of 100 turns and radius 0.01 m is placed with its axis coinciding with the solenoid axis. The current in the solenoid reduces at a constant rate to 0 A from 4 A in 0.05 s. If the resistance of the coil is $10\pi^2\Omega$, the total charge flowing through the coil during this time is: (2017-Delhi)

a. 16 μC b. 32 μC c. $16\pi \mu\text{C}$ d. $32\pi \mu\text{C}$

Answer Key

1. (b) 2. (b) 3. (a) 4. (c) 5. (a) 6. (d) 7. (b) 8. (c) 9. (d) 10. (c)
11. (d) 12. (d) 13. (d) 14. (b) 15. (d) 16. (d) 17. (a) 18. (d) 19. (d) 20. (c)
21. (b)